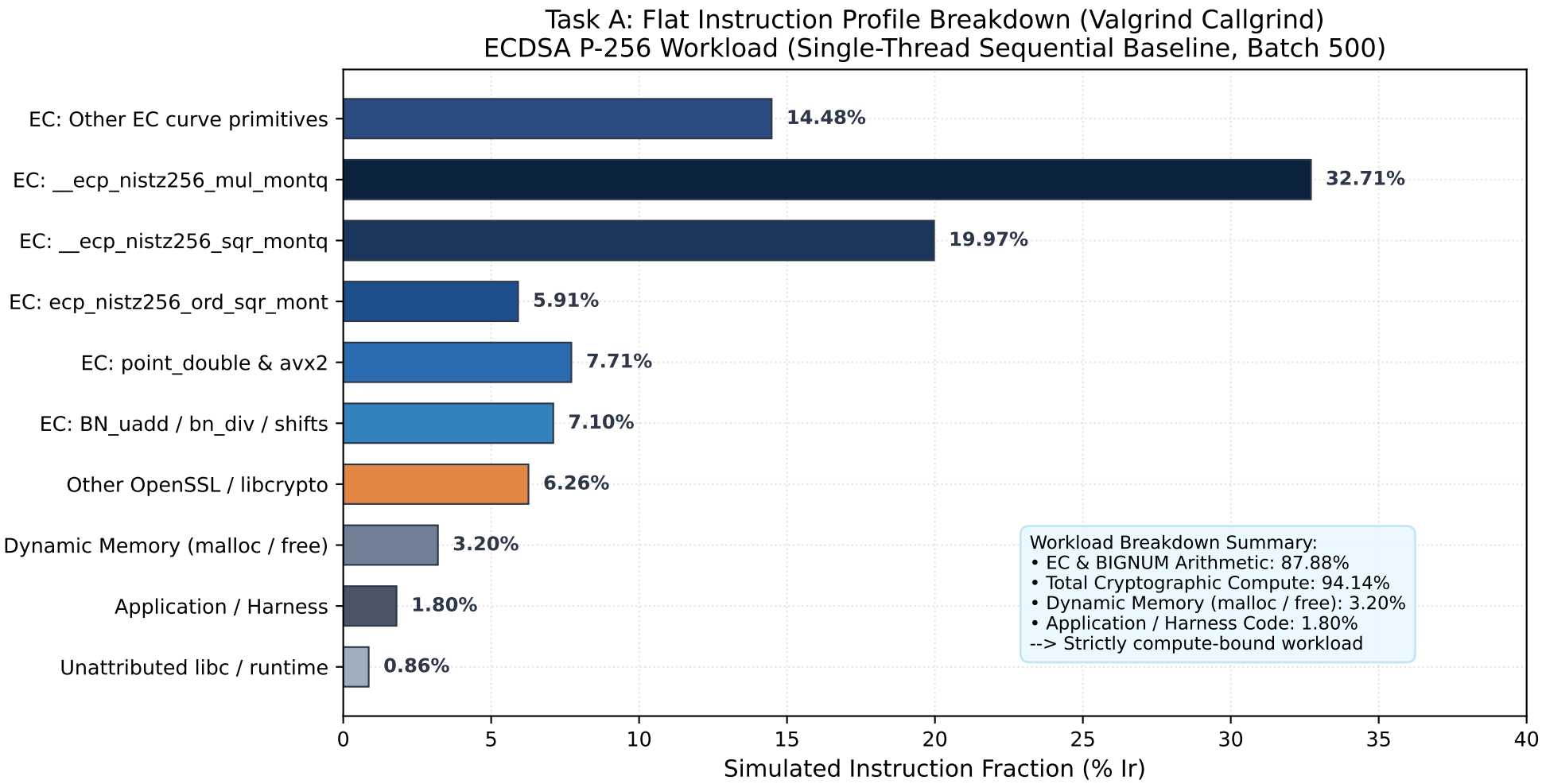
Task A: Flat Instruction Profile Breakdown (Valgrind Callgrind)
ECDSA P-256 Workload (Single-Thread Sequential Baseline, Batch 500)
EC: Other EC curve primitives
EC: __ecp_nistz256_mul_montq
EC: __ecp_nistz256_sqr_montq
EC: ecp_nistz256_ord_sqr_mont
EC: point_double & avx2
EC: BN_uadd / bn_div / shifts
Other OpenSSL / libcrypto
Dynamic Memory (malloc / free)
Application / Harness
Unattributed libc / runtime
14.48%
32.71%
19.97%
5.91%
7.71%
7.10%
6.26%
3.20%
1.80%
0.86%
Workload Breakdown Summary:
• EC & BIGNUM Arithmetic: 87.88%
• Total Cryptographic Compute: 94.14%
• Dynamic Memory (malloc / free): 3.20%
• Application / Harness Code: 1.80%
--> Strictly compute-bound workload
0
5
10
15
20
25
30
35
40
Simulated Instruction Fraction (% Ir)